

LAINGSBURG, South Africa

WMO: 687230

Lat: 33.1908S

Lon: 20.8625E

Elev: 656

StdP: 93.69

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.8	2.0	-8.2	2.0	18.0	-6.2	2.4	16.9	10.6	13.8	9.6	13.6	0.6	330	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	36.0	17.6	34.2	17.6	32.7	17.2	20.5	28.8	19.7	28.6	19.1	28.1	4.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	14.0	23.6	16.9	13.0	23.7	15.9	12.2	23.4	62.0	29.4	59.4	28.3	57.2	28.0	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	7.6	6.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	25.4	25.4	23.1	18.8	14.4	10.8	10.2	12.0	14.7	19.1	21.2	23.8
	DBStd	6.27	3.25	3.00	3.56	3.62	2.68	2.49	2.74	2.63	3.00	3.95	3.75	3.18
	HDD10.0	60	0	0	0	0	2	17	31	9	2	0	0	0
	HDD18.3	1012	1	0	5	39	126	226	251	196	114	39	15	2
	CDD10.0	3057	477	431	406	263	137	42	38	72	144	282	337	428
	CDD18.3	968	219	198	153	52	3	1	0	1	6	62	102	172
	CDH23.3	8412	2051	1757	1221	388	51	8	9	19	74	546	834	1453
	CDH26.7	3521	955	799	488	123	7	0	0	3	16	202	315	614
Wind	WSAvg	2.7	3.2	2.9	2.7	2.4	2.1	2.2	2.2	2.5	2.8	3.1	3.2	3.1
Precipitation	PrecAvg	170	12	14	16	18	17	14	13	5	12	14	14	17
	PrecMax	259	72	50	58	87	55	43	38	35	19	60	60	73
	PrecMin	112	0	0	0	0	0	0	2	0	0	0	0	0
	PrecStd	38	17	14	14	22	14	12	10	4	15	15	15	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.3	37.8	36.0	32.6	27.2	24.8	24.7	26.5	29.0	35.4	35.3	37.6
		MCWB	18.1	17.9	17.4	16.5	13.9	13.3	12.5	11.8	12.9	16.7	16.8	17.5
	2%	DB	35.7	35.2	33.4	30.0	24.6	20.9	21.5	22.5	25.4	31.5	32.3	34.4
		MCWB	17.7	17.7	17.9	16.2	13.3	11.0	10.8	11.2	12.3	15.2	15.8	17.4
	5%	DB	33.7	33.2	31.4	27.7	22.5	18.2	18.7	20.1	23.0	28.7	30.3	32.3
		MCWB	17.7	17.7	17.3	15.9	12.4	10.6	10.1	10.7	11.9	14.4	15.5	17.1
	10%	DB	31.7	31.4	29.5	25.3	20.4	16.1	16.5	17.9	20.8	26.3	28.2	30.2
		MCWB	17.4	17.6	17.1	15.2	12.0	9.9	9.4	9.7	11.1	13.9	15.2	16.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	21.2	21.2	19.1	15.8	13.9	12.7	14.2	16.3	18.1	19.7	20.1
		MCDB	29.9	28.4	29.0	27.3	22.1	20.0	22.5	21.2	21.4	29.1	29.0	29.7
	2%	WB	20.2	20.5	19.9	17.8	14.7	12.8	11.6	12.9	13.9	16.9	18.2	19.3
		MCDB	29.5	28.4	28.1	25.0	21.2	17.5	18.8	19.9	22.4	27.6	26.5	28.9
	5%	WB	19.4	19.8	19.1	17.0	13.7	11.8	10.8	11.8	12.6	15.8	17.1	18.6
		MCDB	29.0	28.3	27.4	23.8	20.6	17.3	16.8	18.0	21.1	25.7	26.7	28.3
	10%	WB	18.7	19.1	18.2	16.1	12.8	10.7	10.0	10.5	11.6	14.9	16.1	17.8
		MCDB	28.6	27.5	26.7	23.7	19.3	15.8	15.5	17.3	19.7	24.4	25.7	27.3
Mean Daily Temperature Range	5% DB	MDBR	12.1	11.7	11.0	10.9	11.2	11.1	12.1	11.5	11.2	11.9	11.9	12.0
		MCDBR	14.5	13.9	13.6	14.3	15.8	15.3	16.8	15.9	15.1	15.8	15.0	14.6
		MCWBR	3.9	3.8	3.7	4.9	6.7	7.8	8.2	7.3	6.4	5.3	4.4	3.7
	5% WB	MCDBR	12.4	11.4	11.3	11.1	12.5	12.7	13.6	13.2	13.5	13.6	12.7	12.1
		MCWBR	3.9	3.5	3.7	4.2	5.8	6.8	7.0	6.5	6.0	4.8	4.4	3.5
Clear-Sky Solar Irradiance	taub		0.327	0.328	0.320	0.298	0.283	0.272	0.275	0.298	0.312	0.331	0.316	0.324
	taud		2.544	2.544	2.559	2.626	2.649	2.651	2.632	2.536	2.469	2.446	2.539	2.545
	Ebn at Noon		1011	990	963	934	899	887	900	918	955	975	1015	1017
	Edh at Noon		109	105	97	81	70	65	70	86	104	115	109	110
All-Sky Solar Radiation	RadAvg		8.27	7.42	6.05	4.50	3.34	2.84	3.15	4.00	5.44	6.78	7.86	8.44
	RadStd		0.24	0.21	0.23	0.22	0.17	0.18	0.18	0.18	0.26	0.30	0.29	0.38

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	+1.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page